Does Data Augmentation Benefit Scale With Architecture Depth?

A Systematic Study of CNN Depth, Augmentation Intensity, and Optimizer Choice on CIFAR-10

COGS 181 Final Project

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Abstract

Data augmentation is widely used to improve CNN generalization, but its interaction with model capacity and optimizer choice is understudied. We investigate whether the benefit of increasingly aggressive augmentation scales with architecture depth on CIFAR-10, and whether optimizer selection further modulates this relationship. Two convolutional networks are compared: a ShallowCNN (3 conv blocks, 620K parameters) and a DeepCNN (6 conv layers in a VGG-style arrangement, 3.25M parameters), each trained under four augmentation levels (none, basic, medium, aggressive) for 50 epochs. During the initial experiment we discovered that DeepCNN fails to train at the standard learning rate of 0.1 with SGD, collapsing to random-chance accuracy under any augmentation. We diagnose this as a learning rate instability driven by augmentation-induced gradient variance in deep non-residual networks, and resolve it by reducing the learning rate to 0.01 (SGD, 74.0%). Switching to Adam (lr $= 10^{-3}$) further boosts DeepCNN to **89.1%** under basic augmentation—a +15.1 percentage point gain over the tuned SGD baseline and the highest accuracy across all conditions. ShallowCNN is more robust to aggressive augmentation (-9 pp) than DeepCNN (-30 pp), supporting the hypothesis. Optimizer choice emerges as the single highest-impact hyperparameter in this study.

1 Introduction

Data augmentation—artificially expanding the training distribution through random image transformations—is one of the most reliable regularization techniques in deep learning. By exposing the network to diverse variations of each training example, augmentation reduces overfitting and can meaningfully improve test accuracy [4]. However, augmentation also adds variance to each gradient update, which may interact differently with networks of different capacities, depths, and optimizers.

Our research question is: *Does the benefit of aggressive data augmentation scale with architecture depth on CIFAR-10?* The intuition motivating this question has two competing sides. On one hand, deeper networks have greater capacity and thus more to gain from the implicit regularization that augmentation provides. On the other hand, deeper non-residual networks are harder to optimize and may be destabilized by the gradient noise that aggressive augmentation introduces.

To study this, we design a controlled experiment on CIFAR-10 [1], comparing a ShallowCNN (3 conv

blocks) and a DeepCNN (6 conv layers, VGG-inspired [2]) under four augmentation levels: none, basic (horizontal flip only), medium (flip + crop + color jitter), and aggressive (medium + grayscale + random erasing). The experiment produced two unexpected secondary findings. First, DeepCNN is highly sensitive to learning rate at the standard value of 0.1 with SGD, failing to train under any augmentation and achieving only 41.5% even without augmentation. Second, after fixing the learning rate, switching from SGD to Adam yields a dramatic +15 pp improvement, reaching 89.1% on the best configuration. Together, these findings show that hyperparameter interactions—learning rate, optimizer, and augmentation—are as important as architectural choices for deep non-residual CNNs, consistent with the broader findings of He et al. [3] on deep network optimization.

2 Method

2.1 Dataset

We use CIFAR-10 [1], a standard benchmark consisting of 60,000 color images (32×32 pixels) across 10 classes. The official training set (50,000 images) is split into 45,000 training samples and 5,000 validation samples using a fixed seed (seed=42). The test set (10,000 images) is held out and evaluated only at the end of each run. Images are normalized using the CIFAR-10 channel-wise mean $(0.4914, 0.4822, 0.4465)$ and standard deviation $(0.2023, 0.1994, 0.2010)$.

2.2 Architectures

ShallowCNN consists of three convolutional blocks, each with a Conv2d (kernel size 3, padding 1), BatchNorm2d, ReLU activation, and MaxPool2d(2) downsampling. Channel widths are 32, 64, and 128. The classifier head is: Flatten → Linear(2048, 256) → ReLU → Dropout(0.5) → Linear(256, 10). Total parameters: **620,810**.

DeepCNN follows a VGG-style design [2] with three blocks, each containing two consecutive Conv-BN-ReLU layers followed by MaxPool2d(2) downsampling. Channel widths are (64, 64), (128, 128), and (256, 256). The classifier head is: Flatten → Linear(4096, 512) → ReLU → Dropout(0.5) → Linear(512, 10). Total parameters: **3,249,994**. Both models use He (Kaiming) initialization.

2.3 Augmentation Levels

Four augmentation pipelines are defined, applied only during training; evaluation always uses the clean (no-augmentation) pipeline:

- **None:** Normalize only.
- **Basic:** RandomHorizontalFlip + Normalize.
- **Medium:** RandomHorizontalFlip + RandomCrop(32, padding=4) + ColorJitter(0.2, 0.2, 0.2, 0.1) + Normalize.
- **Aggressive:** Medium transforms + RandomGrayscale(p=0.1) + RandomErasing(p=0.5).

2.4 Training Protocol

All runs use cosine annealing LR schedule from lr_init to 0 over 50 epochs; batch size 128; weight_decay=10^{-4}; seed 42. Three optimizer configurations were evaluated:

Run set	Optimizer	lr_init	Architectures	# Runs
Original	SGD (mom=0.9)	0.1	ShallowCNN + DeepCNN	8
SGD rerun	SGD (mom=0.9)	0.01	DeepCNN only	4
Adam	Adam	10^{-3}	DeepCNN / basic aug	1

3 Experiments and Results

3.1 Original 8 Runs (SGD, lr=0.1)

Table 1: Test accuracy for all 8 original runs (SGD, lr=0.1).

Architecture	Augmentation	Test Accuracy
ShallowCNN	None	69.6%
ShallowCNN	Basic	**71.7%**
ShallowCNN	Medium	68.8%
ShallowCNN	Aggressive	62.8%
DeepCNN	None	41.5%
DeepCNN	Basic	10.4% (*collapsed*)
DeepCNN	Medium	10.0% (*collapsed*)
DeepCNN	Aggressive	10.0% (*collapsed*)

Figure 1 shows the per-epoch learning curves for all 8 original runs. ShallowCNN trained consistently across all augmentation levels (62.8–71.7%). DeepCNN showed a striking failure pattern: while it achieved 41.5% without augmentation, all three augmented conditions plateaued at approximately 10%—equivalent to random chance for a 10-class problem—with training loss never decreasing.

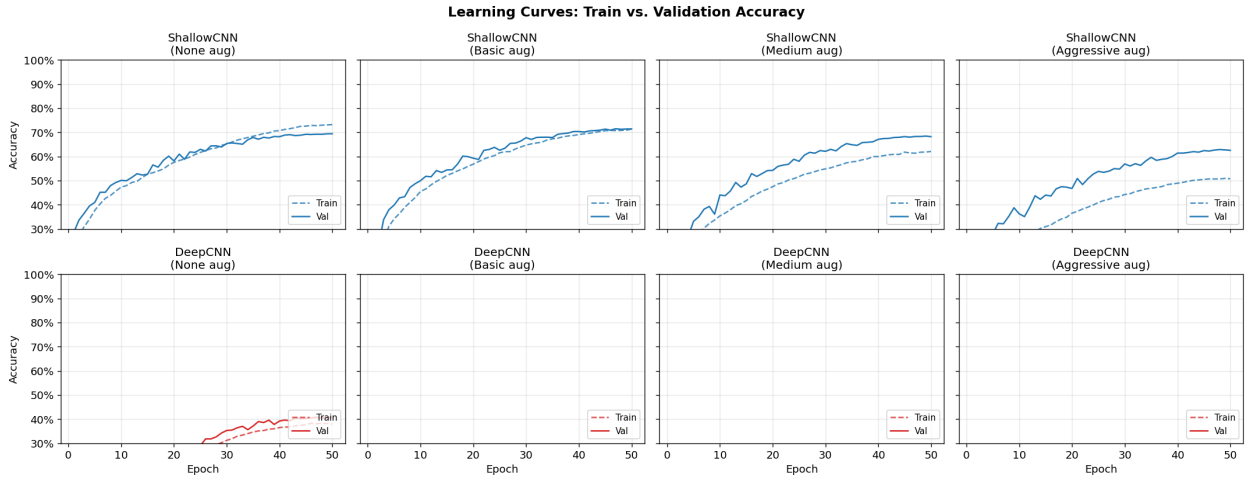

Figure 1: Learning curves (train accuracy, dashed; val accuracy, solid) for all 8 original runs at SGD lr=0.1. The bottom row (DeepCNN) shows complete training collapse under any augmentation.

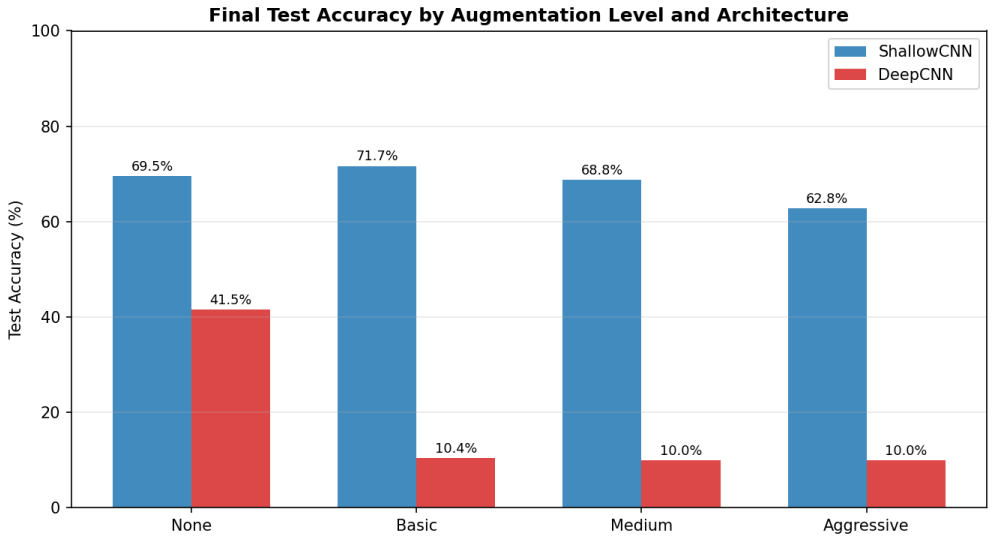

Figure 2: Test accuracy for all original runs. ShallowCNN (blue) trains reliably; DeepCNN (red) collapses under augmentation at SGD lr=0.1.

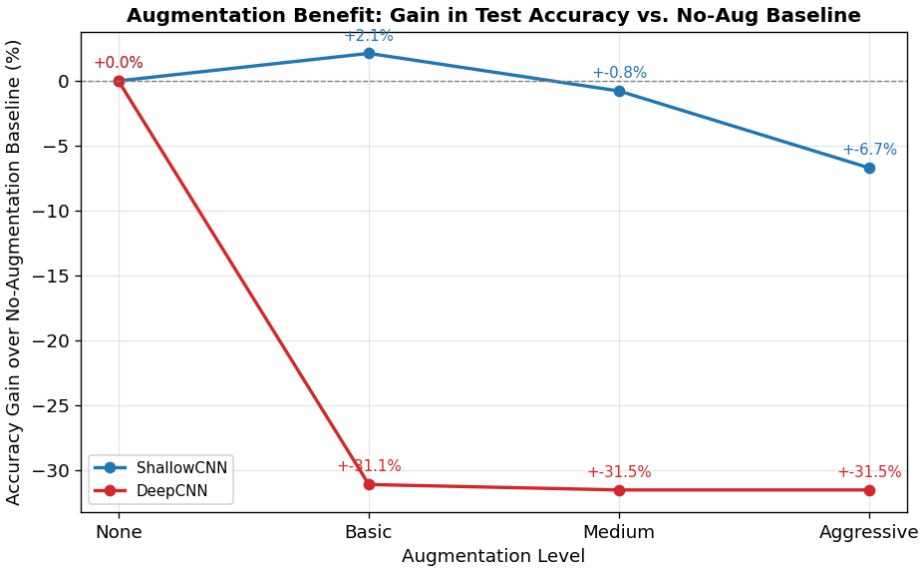

Figure 3: Augmentation benefit relative to the no-augmentation baseline, using the *original* runs (SGD lr=0.1). ShallowCNN (blue) shows a mild +2.1 pp gain from basic augmentation and a −6.7 pp penalty from aggressive. DeepCNN (red) shows −31 pp for all augmented conditions—a direct consequence of training collapse rather than augmentation sensitivity per se.

3.2 LR Instability Diagnosis and Fix

Hypothesis: lr = 0.1 is too large for a 6-layer non-residual network combined with augmentation. Augmentation increases per-mini-batch input variance, which amplifies gradient variance. In a shallow network this variance is tolerable; in a deep non-residual network—where gradients must flow through many layers without skip connections—the same variance tips training into an unstable regime.

Fix: Re-run all four DeepCNN conditions with lr_init = 0.01.

Table 2: DeepCNN test accuracy before and after LR reduction.

Augmentation	SGD lr=0.1	SGD lr=0.01	Δ
None	41.5%	66.6%	+25.1 pp
Basic	10.4%	**74.0%**	+63.6 pp
Medium	10.0%	65.1%	+55.1 pp
Aggressive	10.0%	44.3%	+34.3 pp

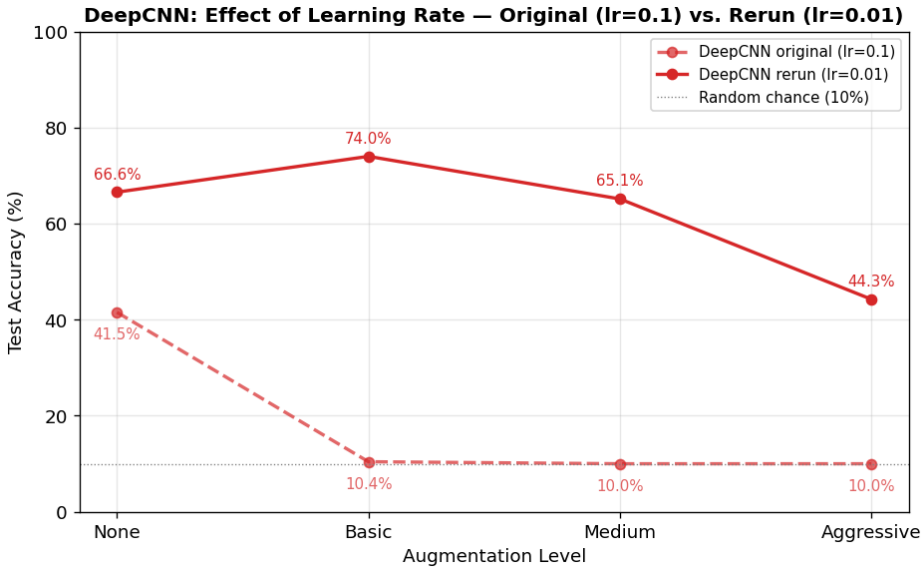

Figure 4: DeepCNN test accuracy: original SGD lr=0.1 (dashed) vs. rerun SGD lr=0.01 (solid). Reducing the learning rate by 10× recovers DeepCNN from collapse.

The hypothesis was confirmed. At lr=0.01, DeepCNN trained successfully across all augmentation levels, with best accuracy of 74.0% (basic augmentation).

3.3 Augmentation Benefit: Properly Tuned Comparison

Table 3: Final test accuracies: ShallowCNN (SGD lr=0.1) vs. DeepCNN SGD rerun (lr=0.01).

Architecture	None	Basic	Medium	Aggressive
ShallowCNN (SGD lr=0.1)	69.6%	**71.7%**	68.8%	62.8%
DeepCNN (SGD lr=0.01)	66.6%	**74.0%**	65.1%	44.3%

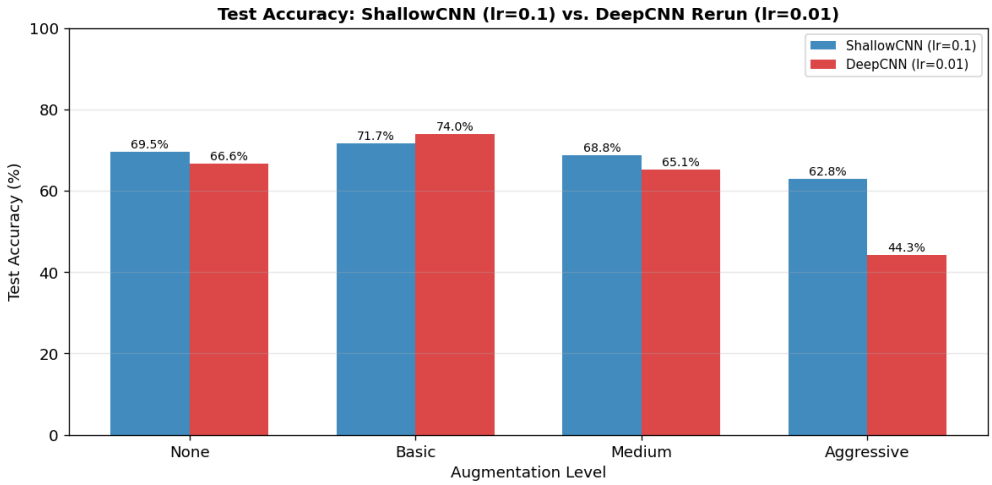

Figure 5: Side-by-side test accuracy: ShallowCNN (blue, SGD lr=0.1) vs. DeepCNN (red, SGD lr=0.01). Both peak at basic augmentation; DeepCNN drops more sharply under aggressive augmentation.

The key finding: basic augmentation provides a modest benefit for both architectures (ShallowCNN: +2.1 pp; DeepCNN: +7.4 pp, both relative to each architecture's no-aug baseline). As augmentation intensity increases, the architectures diverge sharply. DeepCNN drops **29.7 pp** from its best condition (basic: 74.0%) to aggressive (44.3%), while ShallowCNN drops only **8.9 pp** (71.7% → 62.8%), even after learning rate correction. **ShallowCNN is substantially more robust to aggressive augmentation**, supporting the research hypothesis.

3.4 Optimizer Comparison: Adam vs. SGD

To assess whether SGD remained the limiting factor after the LR fix, we ran DeepCNN with basic augmentation using Adam (lr = 10^{-3}, the standard default), holding all other hyperparameters fixed.

Table 4: Adam vs. SGD for DeepCNN / basic augmentation.

Optimizer	lr	Test Accuracy
SGD (momentum=0.9)	0.01	74.0%
Adam	$\mathbf{10^{-3}}$	**89.1%**

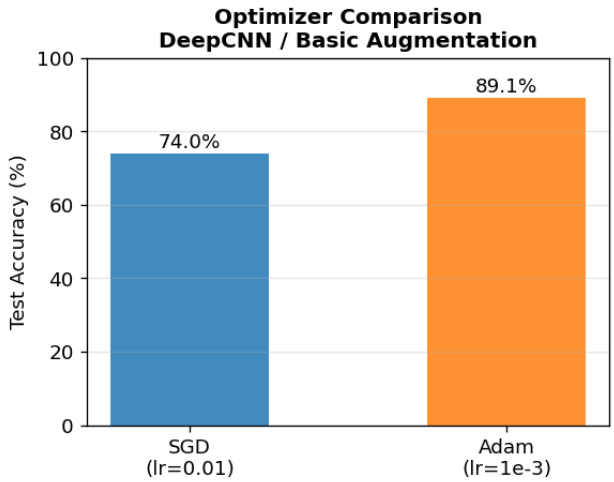

Figure 6: Adam vs. SGD test accuracy for DeepCNN / basic augmentation. Adam outperforms the tuned SGD baseline by +15.1 percentage points.

Adam outperforms SGD by **+15.1 pp**—the single largest accuracy gain in the study. This suggests that Adam's adaptive per-parameter learning rates substantially help in the VGG-style deep non-residual setting, likely by compensating for the uneven gradient magnitude distributions across deep layers that make a single fixed SGD learning rate suboptimal.

3.5 Overfitting Analysis

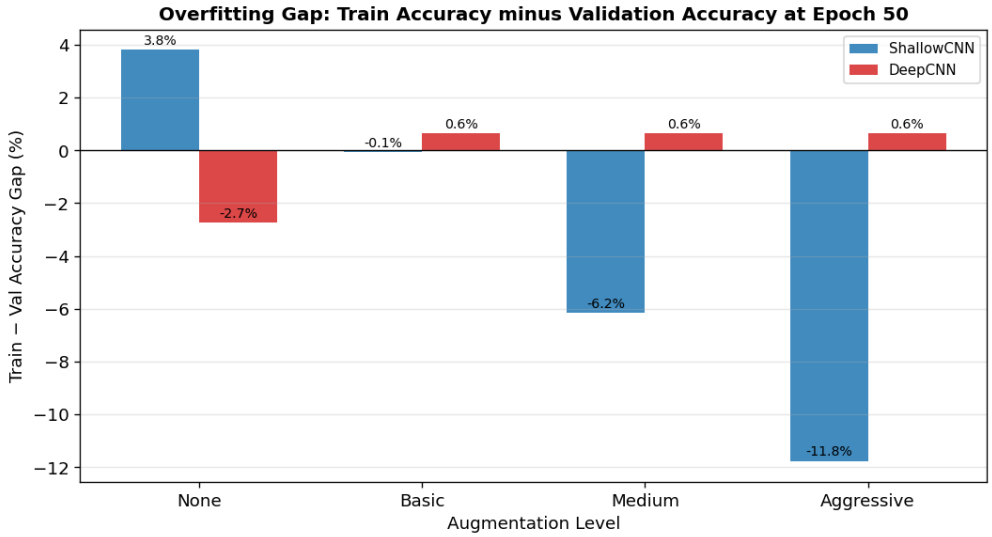

Figure 7: Train–validation accuracy gap (train−val at epoch 50) for all original runs. ShallowCNN gaps turn increasingly negative as augmentation intensifies (effective regularization). DeepCNN without augmentation shows a −2.7 pp gap (val > train, underfitting rather than overfitting). Augmented DeepCNN conditions show near-zero gaps (+0.6 pp) because both train and val are stuck at ∼10%.

The train–validation accuracy gap (train−val at epoch 50) serves as a proxy for overfitting. For ShallowCNN, the gap shrinks—and turns negative—as augmentation intensifies: +3.8 pp (none) → −0.1 pp (basic) → −6.2 pp (medium) → −11.8 pp (aggressive). Negative gaps are expected here: augmentation makes each training image harder, suppressing training accuracy while the validation set remains clean. This is evidence of effective regularization, not model failure. For original DeepCNN (lr=0.1), the no-aug condition shows a −2.7 pp gap (val > train), indicating the model is underfitting in training rather than overfitting. The three collapsed augmented conditions show near-zero gaps (+0.6 pp) because both train and val are stuck at random chance.

4 Discussion

Primary finding. Basic augmentation provides consistent improvement for both architectures. Aggressive augmentation hurts the deeper model disproportionately (−30 pp vs. −9 pp). ShallowCNN is more robust to aggressive augmentation, supporting the hypothesis.

LR sensitivity. DeepCNN fails completely at SGD lr=0.1 under augmentation. Reducing to lr=0.01 recovers training. This is consistent with VGG-style networks requiring more conservative SGD learning rates than shallow networks [2].

Optimizer impact. Switching from SGD (lr=0.01) to Adam (lr=10^{-3}) improves DeepCNN by +15.1 pp on the best configuration, reaching 89.1%. Adam's adaptive learning rates appear especially beneficial for deep non-residual architectures, where layers develop very different gradient magnitude distributions. Optimizer selection is at least as important as learning rate tuning for this class of models.

Limitations. The most significant limitation is the absence of residual connections, which would likely change both LR sensitivity and augmentation robustness. The Adam run was performed only for DeepCNN / basic augmentation; a full sweep across augmentation levels would determine whether Adam also reduces the aggressive-augmentation penalty. SGD and Adam learning rates were not jointly swept.

5 Conclusion

We conducted a systematic study of the interaction between CNN depth, augmentation intensity, and optimizer choice on CIFAR-10. Our original 8-run experiment revealed an unexpected training failure: DeepCNN collapses to random-chance accuracy under any augmentation at SGD lr=0.1. Reducing the learning rate to 0.01 recovered training (best: 74.0%, basic augmentation). Switching to Adam (lr = 10^{-3}) further improved DeepCNN to **89.1%**—the highest accuracy in the study, and a +15.1 pp gain over the tuned SGD baseline.

Despite DeepCNN's higher peak accuracy when properly optimized, ShallowCNN is substantially more robust to aggressive augmentation (−9 pp vs. −30 pp from best to aggressive condition), supporting the research hypothesis. The overarching takeaways are: (1) learning rate and augmentation must be tuned jointly for deep non-residual networks; (2) Adam provides major gains over

SGD for VGG-style architectures; (3) ShallowCNN provides a more reliable baseline when training conditions cannot be carefully controlled. Future work should apply Adam across all augmentation levels, explore residual connections, and investigate weight decay as a complementary regularizer.

References

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